KAIZEN MATHS

Lesson starter

Trig Identities, Proofs and Equations

Equations

Name _____

S1

Solve $3\operatorname{cosec}\theta = 8$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

S2

Solve, for $-180° \leq x \leq 180°$, the equation $4\cos x + \cot x = 0$. Give answers to 1 decimal place.

S3

Solve, for $0 \leq x \leq 360°$, $\cot^2 x = 1$. Give answers to 1 decimal place.

S4

Solve, for $-180° \leq \theta \leq 180°$, $\sec(2\theta + 10°) = -1.3$. Give answers to 1 decimal place.

Show your working

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ALG-44E5-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Trig Identities, Proofs and Equations

Equations

Name _____

E1

Solve $3\operatorname{cosec}\theta = 7$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

E2

Solve, for $0 \leq x \leq 2\pi$, the equation $2\cot x = 3\sec x$. Give your answers in terms of π.

E3

Solve, for $0 \leq x \leq 360°$, $\sec^2 x - 7\sec x + 10 = 0$. Give answers to 1 decimal place.

KAIZEN MATHS

Exit ticket

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Equations

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Trig Identities, Proofs and Equations

Equations

Name _____

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Exit ticket

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Teacher answers & working

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S1

Solve $3\operatorname{cosec}\theta = 8$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Answer:

$\theta = 22.0°, 158.0°$

Working:

Divide by the coefficient.

$\operatorname{cosec}\theta = \frac{8}{3}$

Use the reciprocal definition.

$sin\,\theta = 0.3750$

Find every solution in the interval $0 \leq \theta \leq 360°$.

$\theta = 22.0°, 158.0°$

S2

Solve, for $-180° \leq x \leq 180°$, the equation $4\cos x + \cot x = 0$. Give answers to 1 decimal place.

Answer:

$x = -165.5°, -90.0°, -14.5°, 90.0°$

Working:

Replace $\cot x$ with $\frac{\cos x}{\sin x}$.

$4\cos x + \frac{\cos x}{\sin x} = 0$

Multiply by $\sin x$.

$4\sin x \cos x + \cos x = 0$

Factorise.

$\cos x(4\sin x + 1) = 0$

$\cos x = 0$ or $\sin x = -\frac{1}{4}$

$x = -165.5°, -90.0°, -14.5°, 90.0°$

Teacher answers & working

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S3

Solve, for $0 \leq x \leq 360°$, $\cot^2 x = 1$. Give answers to 1 decimal place.

Answer:

$x = 45.0°, 135.0°, 225.0°, 315.0°$

Working:

Square root both sides.

$\cot x = \pm 1$

Use $\cot x = \dfrac{1}{\tan x}$.

$\tan x = \pm 1.0000$

$x = 45.0°, 135.0°, 225.0°, 315.0°$

S4

Solve, for $-180° \leq \theta \leq 180°$, $\sec(2\theta + 10°) = -1.3$. Give answers to 1 decimal place.

Answer:

$\theta = -114.9°, -75.1°, 65.1°, 104.9°$

Working:

Convert from sec to cos.

$\cos(2\theta + 10°) = \dfrac{1}{-1.3} = -0.7692$

Find possible values of $2\theta + 10°$.

$2\theta + 10° = -219.7°, -140.3°, 140.3°, 219.7°$

Subtract $10°$ and divide by 2.

$\theta = -114.9°, -75.1°, 65.1°, 104.9°$

E1

Solve $3 \operatorname{cosec} \theta = 7$ for $0 \leq \theta \leq 360°$. Give answers to 1 decimal place.

Answer:

$\theta = 25.4°, 154.6°$

Working:

Divide by the coefficient.

$\operatorname{cosec} \theta = \dfrac{7}{3}$

Use the reciprocal definition.

$sin\,\theta = 0.4286$

Find every solution in the interval $0 \leq \theta \leq 360°$.

$\theta = 25.4°, 154.6°$

Teacher answers & working

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E2

Solve, for $0 \leq x \leq 2\pi$, the equation $2\cot x = 3\sec x$. Give your answers in terms of π.

Answer:

$x = \frac{\pi}{6}, \frac{5\pi}{6}$

Working:

Rewrite using sine and cosine.

$2\frac{\cos x}{\sin x} = 3\frac{1}{\cos x}$

Multiply by $\sin x \cos x$.

$2\cos^2 x = 3\sin x$

Use $\cos^2 x = 1 - \sin^2 x$.

$2(1 - \sin^2 x) = 3\sin x$

$2\sin^2 x + 3\sin x - 2 = 0$

$(2\sin x - 1)(\sin x + 2) = 0$

Reject $\sin x = -2$ because sine cannot be less than -1.

$\sin x = \frac{1}{2}$, so $x = \frac{\pi}{6}, \frac{5\pi}{6}$.

E3

Solve, for $0 \leq x \leq 360°$, $\sec^2 x - 7\sec x + 10 = 0$. Give answers to 1 decimal place.

Answer:

$x = 60.0°, 78.5°, 281.5°, 300.0°$

Working:

Treat $\sec x$ as the variable.

Let $u = \sec x$, so $u^2 - 7u + 10 = 0$.

Factorise the quadratic.

$(u - 2)(u - 5) = 0$

$\sec x = 2$ or $\sec x = 5$

Convert to cosine using $\sec x = \frac{1}{\cos x}$.

$\cos x = \frac{1}{2}$ or $\cos x = \frac{1}{5}$

$x = 60.0°, 78.5°, 281.5°, 300.0°$